

NODE 1

Penile Tumescence, Rigidity, and Physiology Monitoring Wearable

Vibration, Haptic, and Active Stimulus Integration

Field

This disclosure relates to wearable physiological monitoring devices and, more particularly, to the integration of vibration, haptic, and active mechanical stimulus elements within a penile tumescence and rigidity monitoring wearable.

Overview

In certain embodiments, the device incorporates one or more vibration, haptic, or active mechanical actuation elements. These may operate independently or in coordination with physiological sensing subsystems to provide controlled tactile stimulus, arousal activation, neuromodulatory input, biofeedback, perceptual signaling, or user guidance during wear. The device may be configured as a C-shaped wearable ring or a fully enclosed circumferential ring for extended daytime, nighttime, or sleep use.

Actuator Technologies

The actuation subsystem may include one or more of the following, individually or in combination:

- ERM vibration motors; LRA actuators; piezoelectric vibration or bending actuators
- Electromagnetic or voice-coil haptic actuators; shape-memory alloy micro-actuators
- Electroactive polymer or dielectric elastomer actuators

Actuators may be selected based on size, power efficiency, frequency response, or mechanical isolation requirements.

Actuator Placement and Mechanical Integration

Actuators may be embedded within the wearable structure, including but not limited to:

- Inner circumferential surfaces; discrete circumferential zones (dorsal, ventral, lateral)
- Floating or elastomer-isolated mounts; encapsulated silicone or polymer housings

Mechanical isolation structures may reduce interference with pressure, strain, PPG, and inertial sensors.

Stimulus Characteristics

The vibration or haptic output may be modulated across one or more dimensions, including:

- Frequency (low, mid, and high vibration bands) and amplitude/displacement
- Pulse duration, burst patterns, ramps, or oscillatory waveforms
- Sequential, alternating, or phase-shifted actuation across multiple actuators

Stimulus parameters may be static, user-configured, or dynamically adjusted.

Functional Roles

The actuation subsystem may serve one or more purposes, including:

- Arousal activation or reinforcement; neuromuscular or vascular stimulation
- Perceptual biofeedback correlated with rigidity, pressure, strain, or blood-flow metrics
- Silent user alerts or state notifications; sleep-state or nocturnal event marking

Closed-Loop Operation

In certain embodiments, actuation is controlled in a closed-loop manner using real-time sensor data (including temperature, pressure, strain, photoplethysmography, or inertial measurements) to enable adaptive stimulus profiles responsive to detected state, context, or changes over time.

Safety and Control

The device may incorporate safety mechanisms including maximum intensity limits, duty-cycle restrictions, user override controls, automatic shutoff conditions, or context-aware inhibition to ensure safe and repeatable operation during extended wear.